

# Research Statement

Abdelilah Aitrouga

## Research Interests

---

My long-term research goal is to develop trustworthy machine learning systems that are capable of learning rich representations from complex multimodal data while remaining interpretable, robust, and aligned with human objectives. My research lies at the intersection of artificial intelligence, representation learning, and computational neuroscience, with a particular focus on understanding how neural representations can be leveraged to build more capable and reliable AI systems.

During my PhD at Mohammed VI Polytechnic University (UM6P), I have investigated neural decoding and multimodal representation learning using large-scale neuroimaging datasets. My current research focuses on reconstructing visual information from fMRI by aligning brain activity with latent representations learned by modern vision foundation models. Through this work, I have gained extensive experience in deep learning, transformer architectures, diffusion models, multimodal learning, and large-scale model training.

Beyond neuroscience, I am interested in the broader question of how internal representations emerge, evolve, and can be controlled in increasingly capable AI systems. I believe that understanding representations is fundamental not only for improving model performance but also for achieving interpretability, robustness, and alignment.

## Future Research Vision

---

The recent transition from static language models to autonomous AI agents presents exciting opportunities as well as significant safety challenges. As agents become capable of maintaining long-term memory, interacting with external environments, and cooperating with other agents, ensuring that they remain aligned with human intentions becomes increasingly difficult. These challenges closely match the research vision of the COMPASS group.

If given the opportunity to join the COMPASS group, I would be particularly interested in investigating:

- **Interpretability of autonomous agents:** developing methods to understand how internal representations evolve during long-horizon reasoning and decision making.

- **Representation learning for AI safety:** studying whether latent representations can be used to detect unsafe behaviors, reasoning failures, or distribution shifts before they manifest in model outputs.
- **Memory-aware safe AI agents:** designing mechanisms that allow long-term memory while preserving privacy, contextual integrity, and user control.
- **Evaluation of foundation models and AI agents:** creating benchmarks that measure robustness, contextual reasoning, and alignment under realistic deployment scenarios rather than static benchmark settings.
- **AI for scientific discovery:** exploring how reliable AI agents can assist researchers in generating hypotheses, integrating multimodal scientific evidence, and accelerating discoveries while maintaining transparency and human oversight.

## Why COMPASS

---

I am particularly attracted to the COMPASS group because it combines fundamental machine learning research with practical AI safety challenges. I appreciate the group's emphasis on understanding failure modes, designing principled mitigation strategies, and developing trustworthy AI systems that remain useful in real-world settings. This research philosophy strongly aligns with my own goal of building AI systems that are not only capable but also reliable and beneficial.

Although my previous work has primarily focused on computational neuroscience and multimodal representation learning, I view these experiences as a strong foundation for addressing broader AI safety questions. Developing robust representations, understanding model behavior, and designing scalable learning algorithms are challenges shared across both NeuroAI and AI safety.

## Long-Term Goal

---

In the long term, I aim to become an independent researcher working at the intersection of machine learning, AI safety, and scientific AI. I hope to contribute to developing AI systems that are interpretable, robust, and aligned with human values while enabling scientific discovery across neuroscience and other domains. I believe that the collaborative environment of the COMPASS group, together with its focus on foundational AI safety research, would provide an exceptional environment in which to pursue this vision.